

10 Czerwca 2026

CSAIMBOT



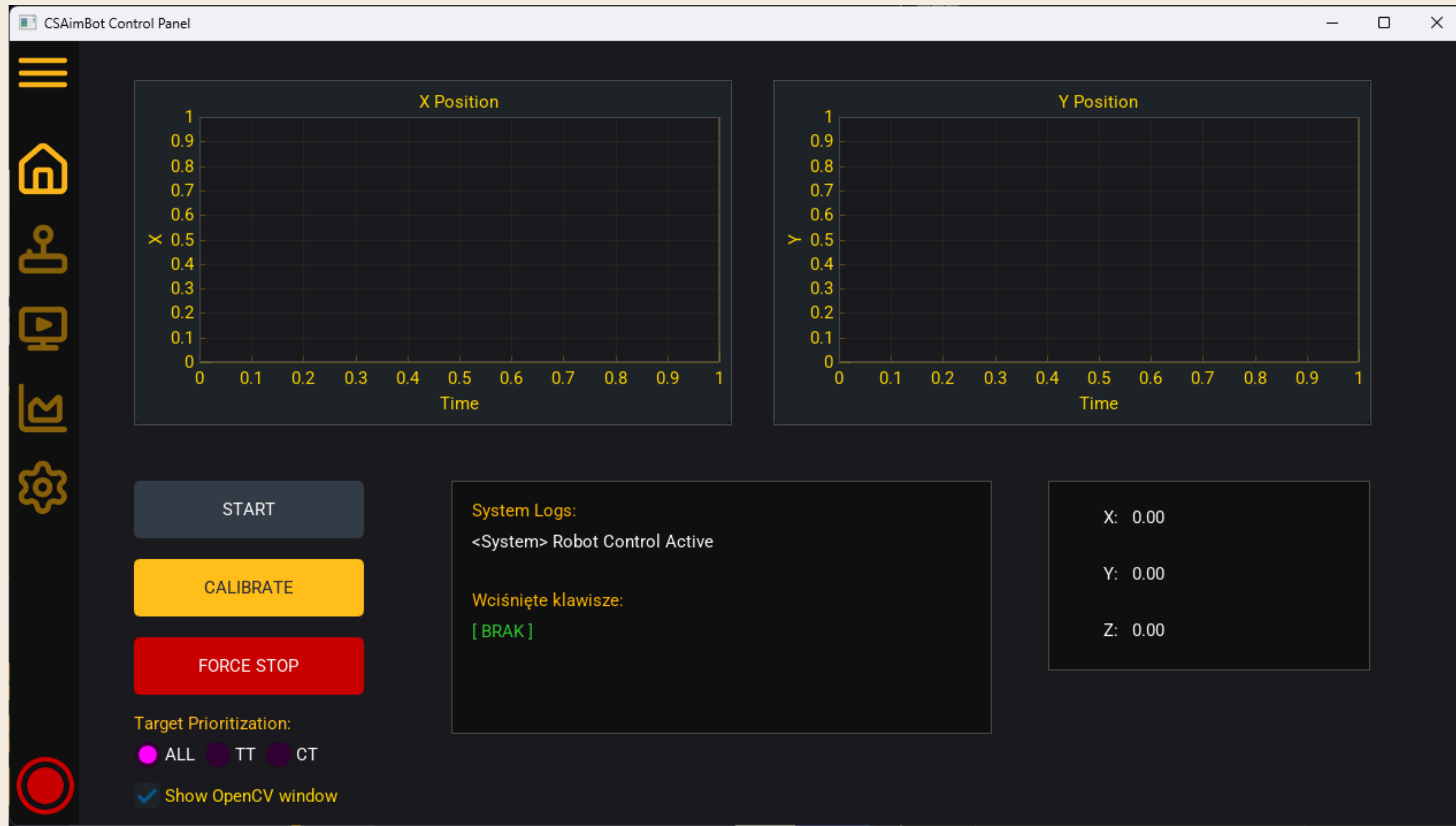
Prezentowane przez
Tomasz Nazar, Filip Pietrzak

CELE PROJEKTU M1+M2+M3

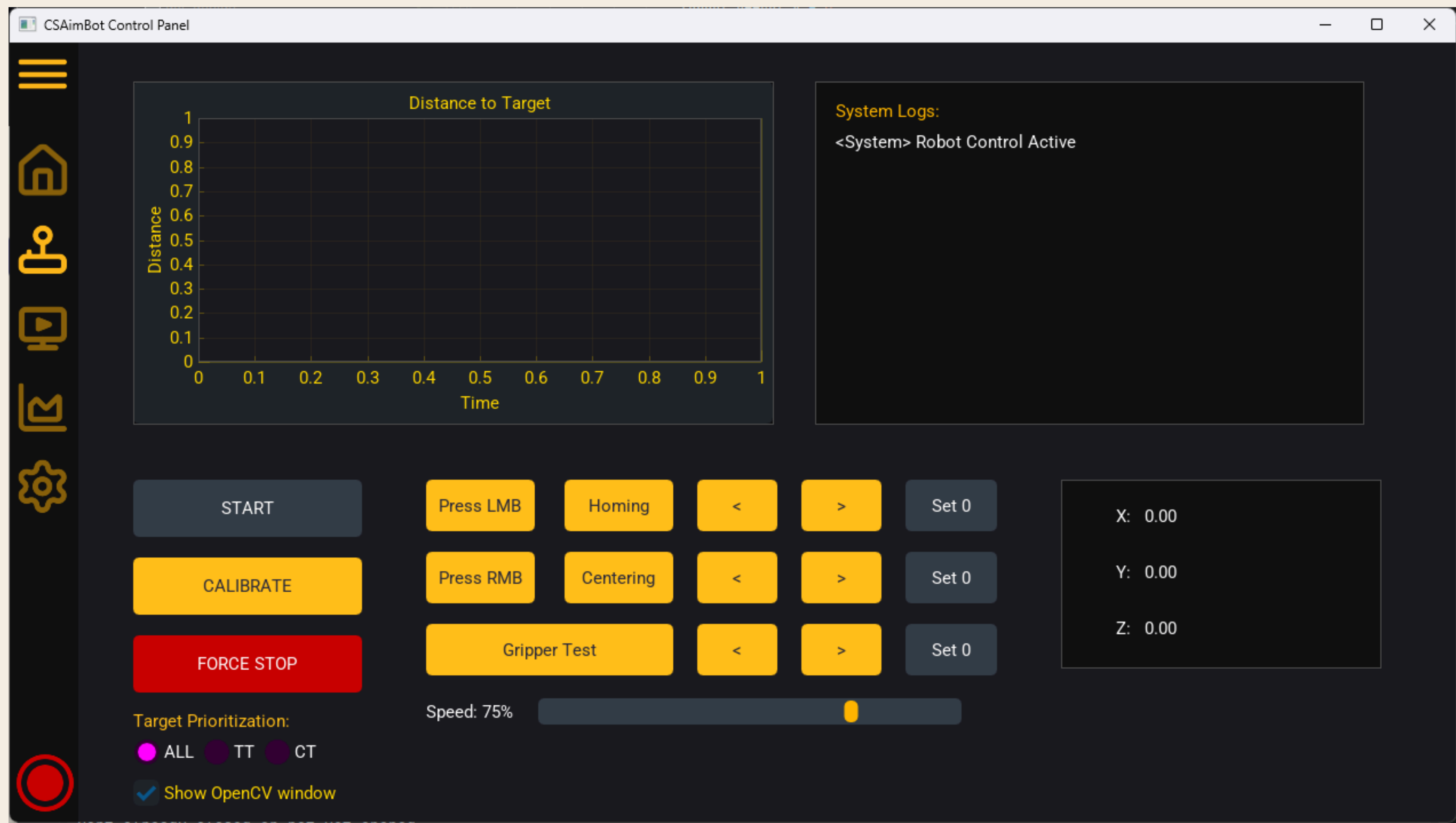
- Integracja całego systemu ✓
- Sterowanie robotem na podstawie poprawnych koordynatów z YOLO ✓
- Poprawnie działająca aplikacja ✓
- Skończenie budowy i projektu robota ✓
- Kompensacja odrzutu broni ✓
- Symulacja w Gazebo ✓
- 60 FPS na RTX 3060 ✓
- Dataset z ponad 10k zdjęć ✓
- Skuteczność YOLO >95% ✓
- Czas reakcji ~10ms ✓
- Dokładność namierzania stojącego celu 100% ✓

OSIĄGNIĘTE REZULTATY

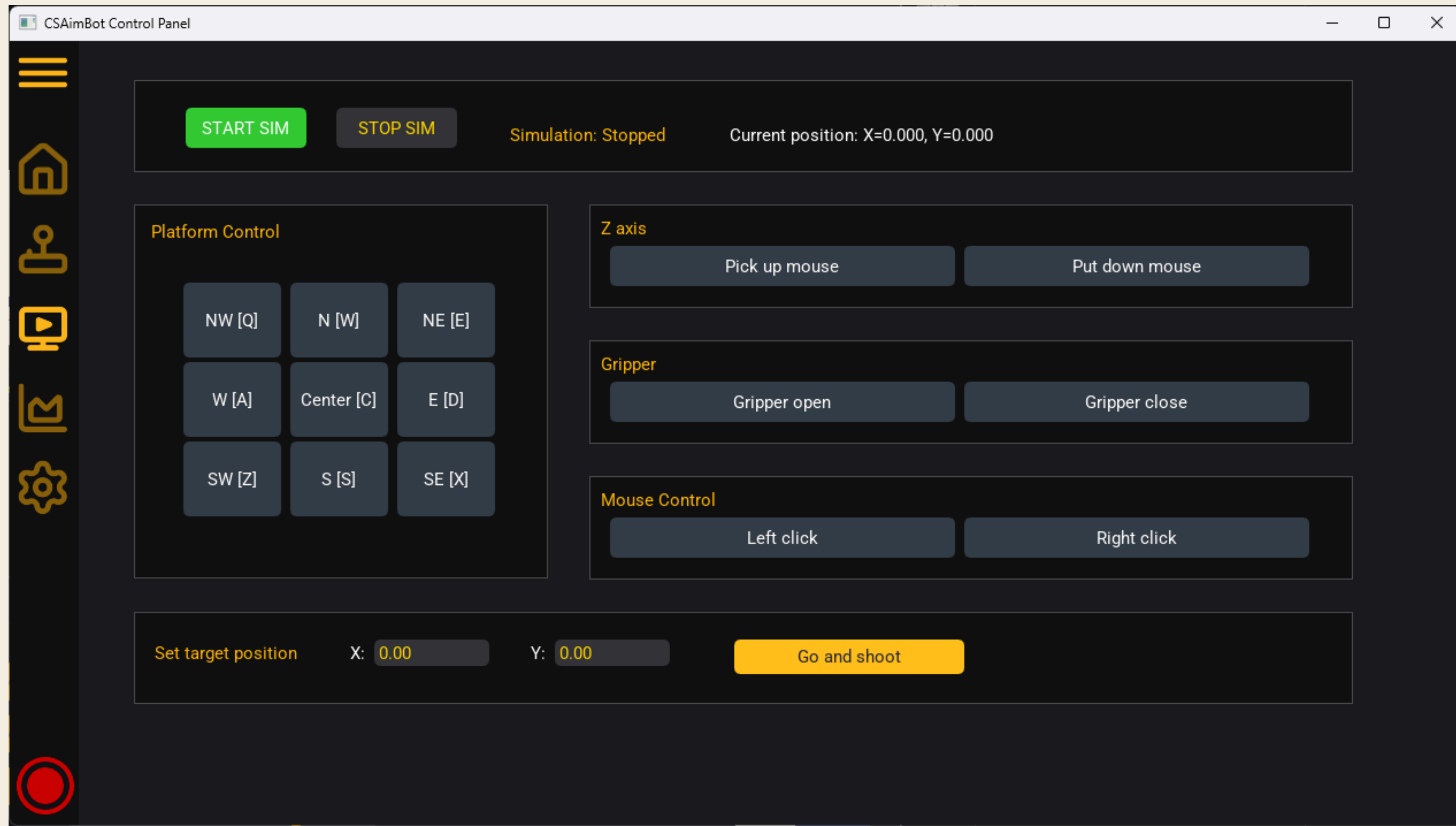
APLIKACJA



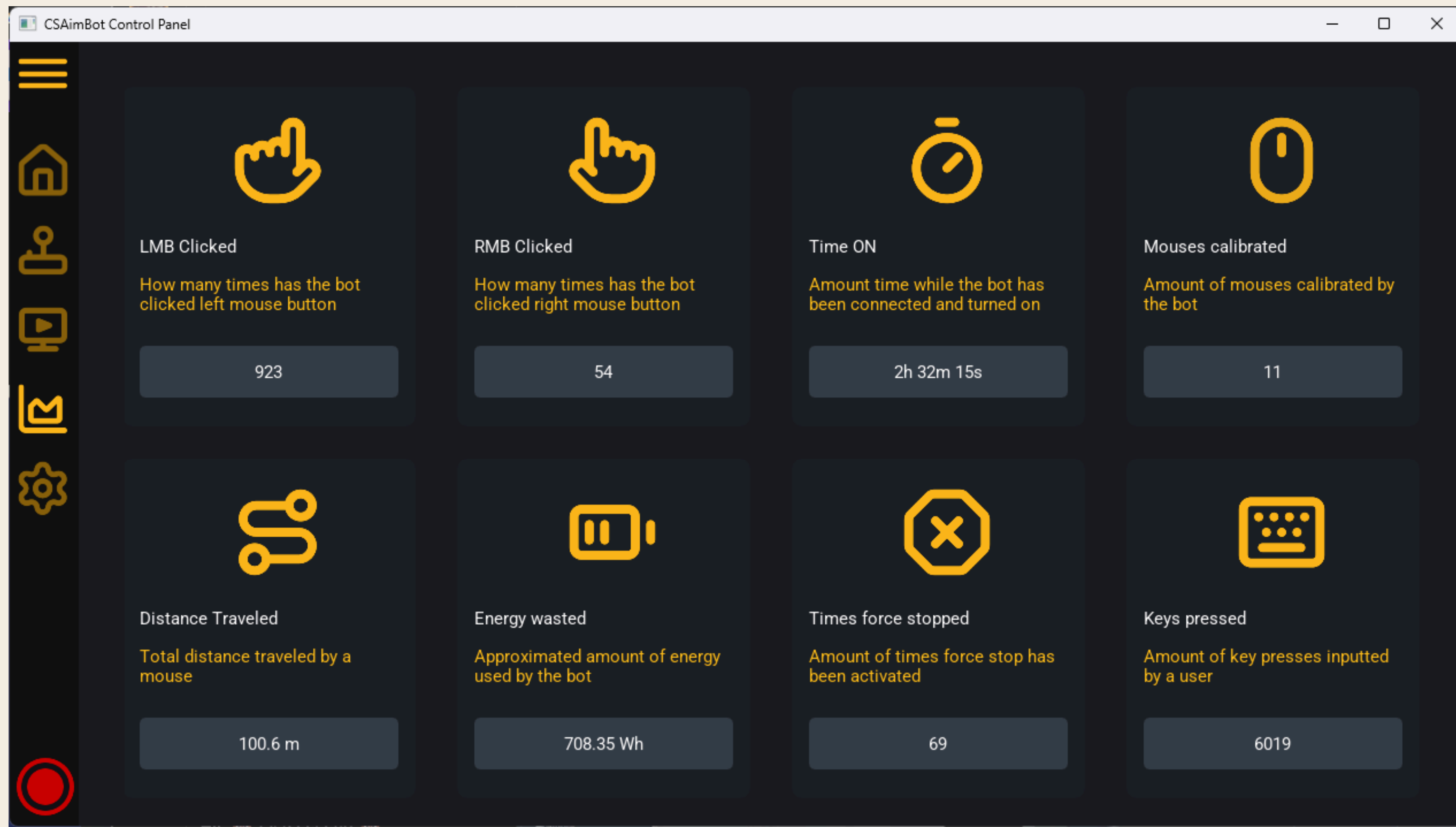
APLIKACJA



APLIKACJA



APLIKACJA



APLIKACJA

CSAimBot Control Panel

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⚙️

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Language: English

Resolution: 1280x720

COM Port: COM3

Kp: 151.0 Ki: 1.65 Kd: 8.85

Deadband: 0.9 Min speed: 0 Accel: 14

Update Settings

INSTALLERY

Udostępnione mnie > CSAimBot installers 

✓ 





Typ elementu ▾

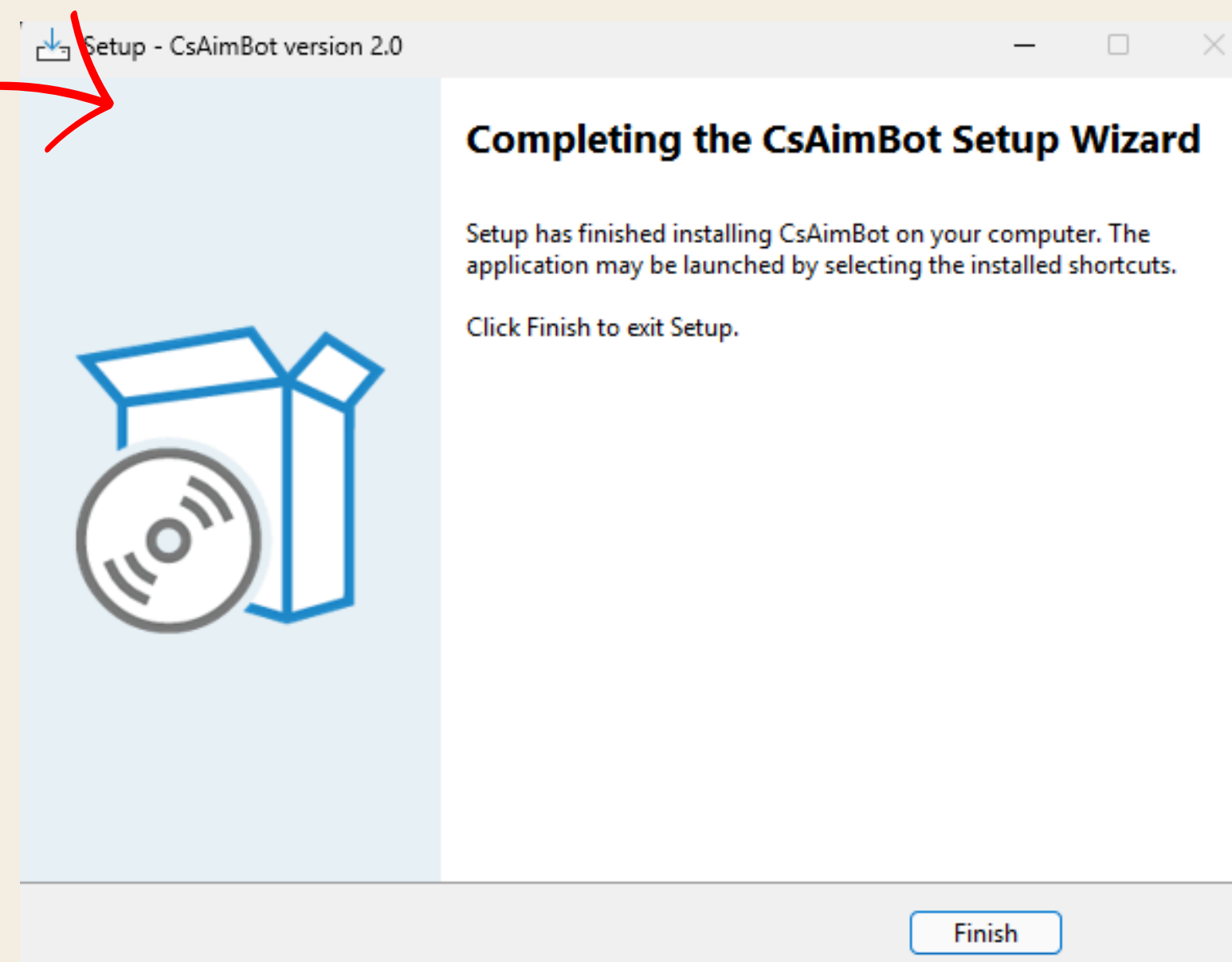
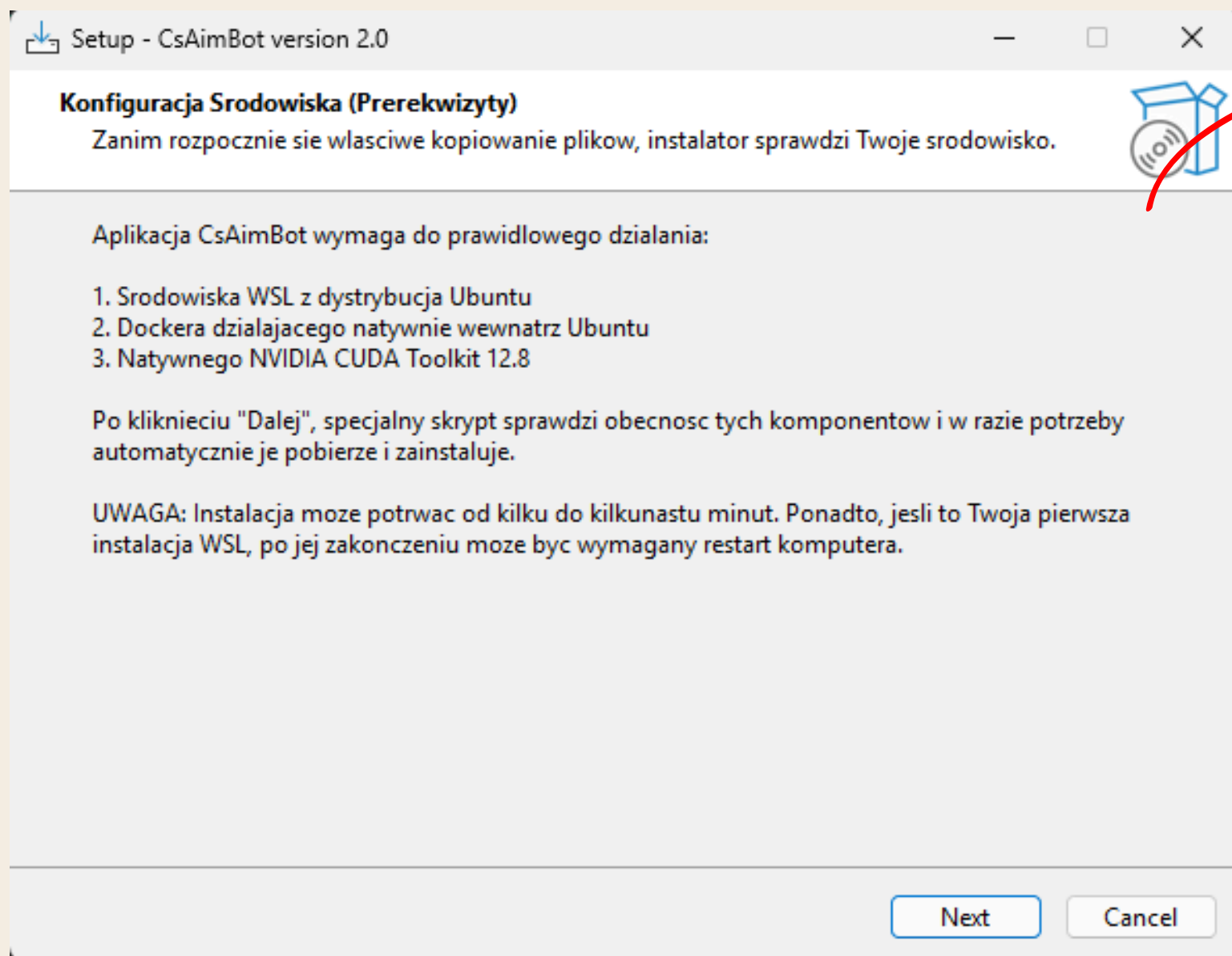
Osoby ▾

Zmodyfikowano ▾

Zródło ▾

Nazwa	Właściciel	Data modyfikacji 	Rozmiar pliku	 Sortuj
 CsAimBot-Installer-x64-v2.run 	 ja	2 cze ja	9,85 GB	
 CsAimBot-Installer-x64-v2-console.zip 	 lukaszmroczek90	19 maj	5,33 GB	

INSTALLER WINDOWS



INSTALLER LINUX

```
karol-puczynski@kontuter:~$ cd Downloads/  
karol-puczynski@kontuter:~/Downloads$ chmod +x CsAimBot-Installer-x64-v2.run  
karol-puczynski@kontuter:~/Downloads$ sudo ./CsAimBot-Installer-x64-v2.run  
[sudo] password for karol-puczynski:  
Verifying archive integrity... 100% MD5 checksums are OK. All good.  
Uncompressing CsAimBot System Installer 100%  
=== Instalator CsAimBot dla Linuxa ===
```

```
[1/4] Sprawdzanie i instalacja Dockera...  
Docker jest już zainstalowany: Docker version 29.5.2, build 79eb04c
```

```
[2/4] Sprawdzanie NVIDIA Container Toolkit...  
NVIDIA Container Toolkit jest już zainstalowany.
```

```
[4/4] Konfiguracja reguł Udev dla urządzeń wejściowych...  
Reguły Udev zostały zaktualizowane.
```

```
=== Kopiowanie plików aplikacji do /opt/CsAimBot ===
```

```
[3/4] Instalowanie natywnego CUDA Toolkit 12.8...  
Instalowanie natywnego CUDA Toolkit 12.8 (Repozytorium Ubuntu)...  
2026-06-09 17:16:21-- https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64/cuda-keyring_1.1-1_all.deb  
Resolving developer.download.nvidia.com (developer.download.nvidia.com)... 95.101.116.86, 95.101.116.81  
Connecting to developer.download.nvidia.com (developer.download.nvidia.com)[95.101.116.86]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 4332 (4.2K) [application/x-deb]  
Saving to: '/tmp/cuda-keyring.deb'  
  
/tmp/cuda-keyring.deb 100%[=====] 4.23K --.-KB/s in 0s  
  
2026-06-09 17:16:22 (614 MB/s) - '/tmp/cuda-keyring.deb' saved [4332/4332]  
  
(Reading database ... 342575 files and directories currently installed.)  
Preparing to unpack /tmp/cuda-keyring.deb ...  
Unpacking cuda-keyring (1.1-1) over (1.1-1) ...  
Setting up cuda-keyring (1.1-1) ...  
Hit:1 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 InRelease  
Hit:2 https://mirror.mwt.me/shiftkey-desktop/deb any InRelease  
Hit:3 https://nvidia.github.io/libnvidia-container/stable/deb/amd64 InRelease  
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease  
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:6 https://dl.google.com/linux/chrome-stable/deb stable InRelease  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [425 kB]  
Hit:8 http://pl.archive.ubuntu.com/ubuntu noble InRelease  
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,763 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [806 kB]  
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,195 kB]  
Get:12 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]  
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [232 kB]  
Hit:14 https://packages.gitlab.com/gitlab/gitlab-ee/ubuntu/noble noble InRelease  
Get:15 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [57.1 kB]  
Get:16 http://pl.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Hit:17 https://cyberbotics.com/debian binary-amd64/ InRelease  
Hit:18 https://deb.opera.com/opera-stable stable InRelease  
Hit:19 http://packages.ros.org/ros2/ubuntu noble InRelease  
Hit:20 http://pl.archive.ubuntu.com/ubuntu noble-backports InRelease  
Get:21 http://pl.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,067 kB]  
Get:22 http://pl.archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [625 kB]  
Get:23 http://pl.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.4 kB]  
Get:24 http://pl.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,699 kB]  
Get:25 http://pl.archive.ubuntu.com/ubuntu noble-updates/universe i386 Packages [1,137 kB]  
Get:26 http://pl.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [331 kB]  
Get:27 http://pl.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.7 kB]  
Fetched 10.7 MB in 8s (1,347 kB/s)  
Reading package lists... Done
```

```
Eksportowanie modelu YOLO do formatu TensorRT (może to potrwać chwilę)...  
Exporting model to TensorRT...  
Exporting YOLO model to TensorRT format...  
Ultralytics 8.4.26 Python-3.13.7 torch-2.10.0+cu128 CUDA:0 (NVIDIA GeForce RTX 3050 Laptop GPU, 3768MiB)  
YOLO26n summary (fused): 122 layers, 2,376,396 parameters, 0 gradients, 5.2 GFLOPs  
  
PyTorch: starting from '/opt/CsAimBot/_internal/application/detection_system/yolo/trained_model.pt' with input shape (1, 3, 736, 1280) BCHW and output shape(s) (1, 300, 6) (5.3 MB)  
  
ONNX: starting export with onnx 1.21.0 opset 18...  
torch/onnx/_internal/torchscript/exporter/symbolic_opset11.py:954: UserWarning: Exporting aten::index operator of advanced indexing in opset 18 is achieved by combination of multiple ONNX operators, including Reshape, Transpose, Concat, and Gather. If indices include negative values, the exported graph will produce incorrect results.  
  return opset9.index(g, self, index)  
ONNX: slimming with onnxslim 0.1.91...  
ONNX: export success 1.0s, saved as '/opt/CsAimBot/_internal/application/detection_system/yolo/trained_model.onnx' (9.6 MB)  
  
TensorRT: starting export with TensorRT 10.15.1.29...  
[06/09/2026-17:17:24] [TRT] [I] [MemUsageChange] Init CUDA: CPU -2, GPU +0, now: CPU 978, GPU 264 (MiB)  
[06/09/2026-17:17:24] [TRT] [I] -----  
[06/09/2026-17:17:24] [TRT] [I] Input filename: /opt/CsAimBot/_internal/application/detection_system/yolo/trained_model.onnx  
[06/09/2026-17:17:24] [TRT] [I] ONNX IR version: 0.0.8  
[06/09/2026-17:17:24] [TRT] [I] Opset version: 18  
[06/09/2026-17:17:24] [TRT] [I] Producer name: pytorch  
[06/09/2026-17:17:24] [TRT] [I] Producer version: 2.10.0  
[06/09/2026-17:17:24] [TRT] [I] Domain:  
[06/09/2026-17:17:24] [TRT] [I] Model version: 0  
[06/09/2026-17:17:24] [TRT] [I] Doc string:  
[06/09/2026-17:17:24] [TRT] [I] -----  
TensorRT: input "images" with shape(1, 3, 736, 1280) DataType.FLOAT  
TensorRT: output "output0" with shape(1, 300, 6) DataType.FLOAT  
TensorRT: building FP16 engine as /opt/CsAimBot/_internal/application/detection_system/yolo/trained_model.engine  
[06/09/2026-17:17:24] [TRT] [I] [MemUsageChange] Init builder kernel library: CPU +252, GPU +2, now: CPU 1406, GPU 266 (MiB)  
[06/09/2026-17:17:24] [TRT] [I] Local timing cache in use. Profiling results in this builder pass will not be stored.
```

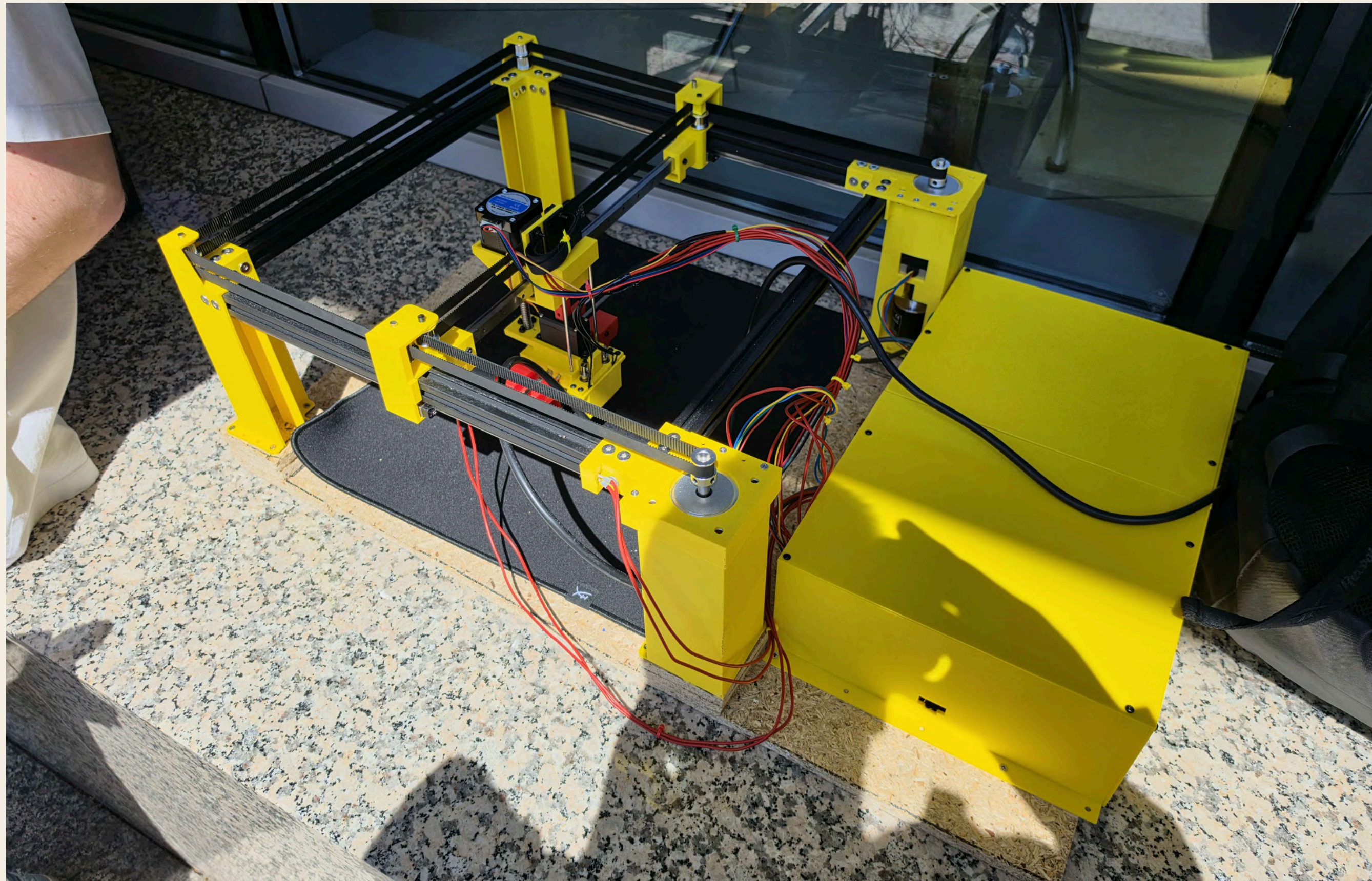
INSTALLER LINUX

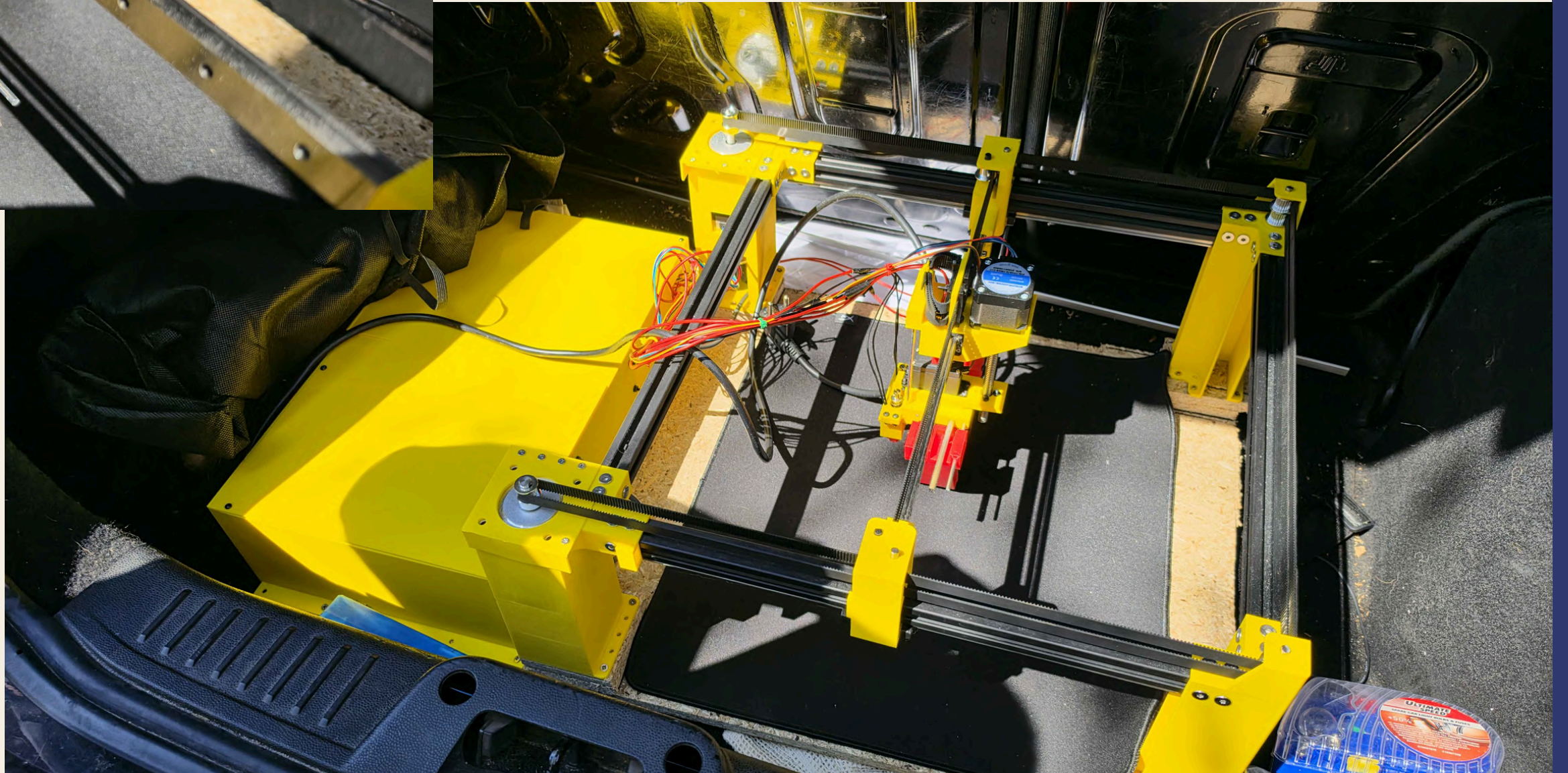
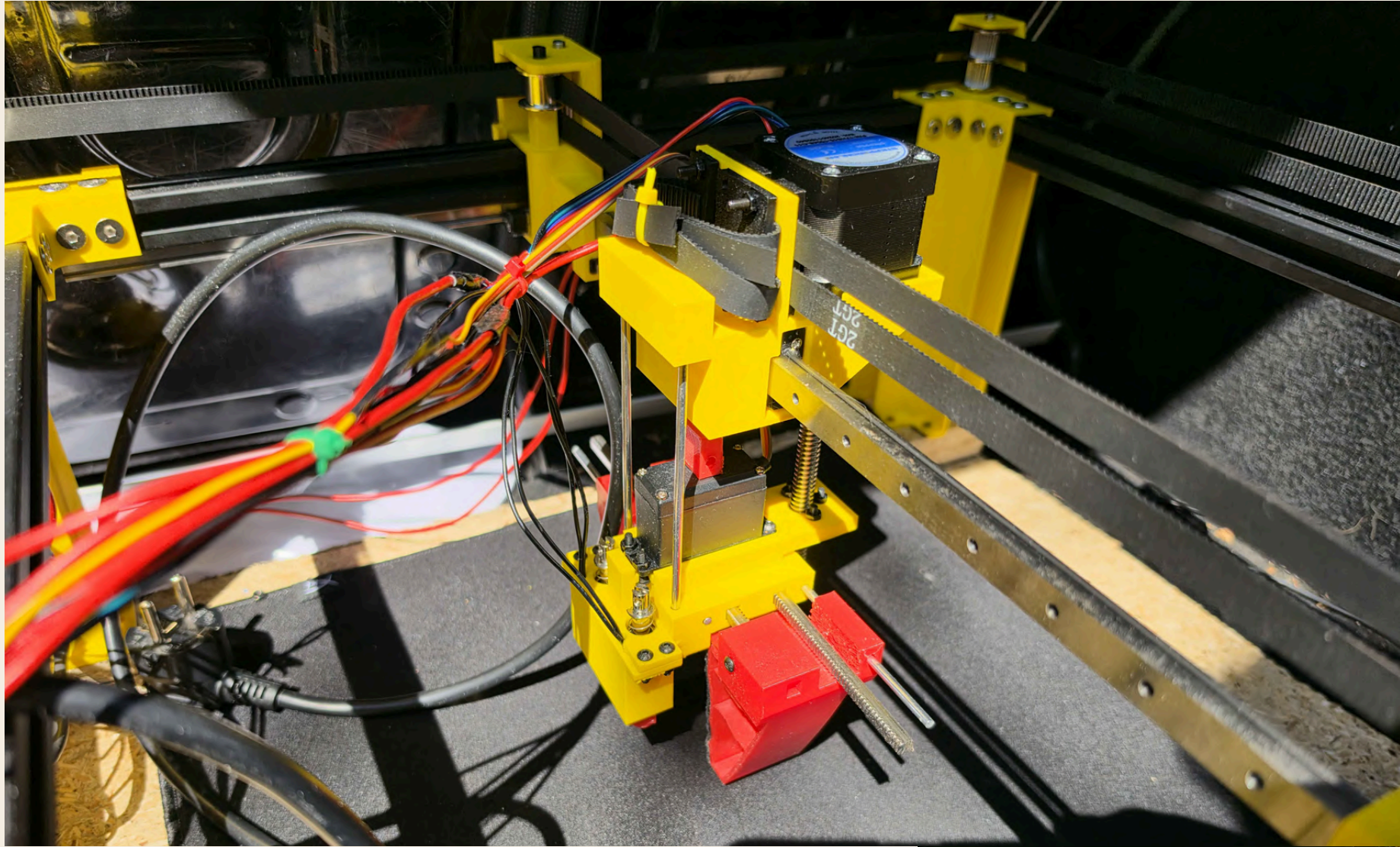
```
Export complete (160.5s)
Results saved to /opt/CsAimBot/_internal/application/detection_system/yolo
Predict:      yolo predict task=detect model=/opt/CsAimBot/_internal/application/detection_
Validate:     yolo val task=detect model=/opt/CsAimBot/_internal/application/detection_syst
lication\detection_system\yolo\training\Beta_v4\Beta_v4_dataset\data.yaml half WARNING ⚠non-P
'imgsz=1280' if val is required.
Visualize:    https://netron.app
Model exported successfully to TensorRT format.

=====
  INSTALACJA ZAKOŃCZONA SUKCESEM!
=====

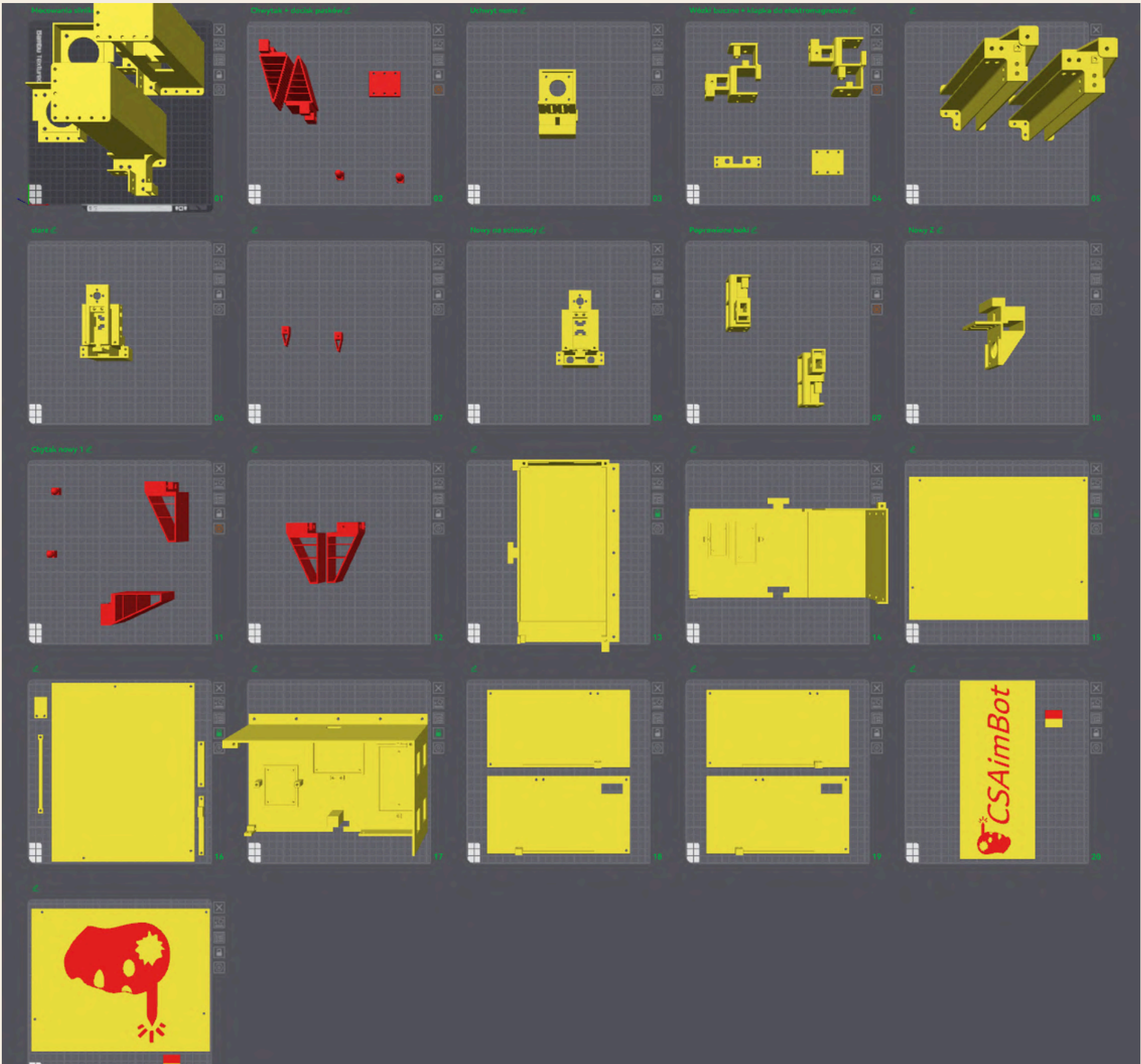
Aplikacja jest dostępna w menu systemowym lub pod ścieżką: /opt/CsAimBot/CsAimBot
Zaleca się restart komputera w celu pełnego zaaplikowania uprawnień Dockera.
```


HARDWARE





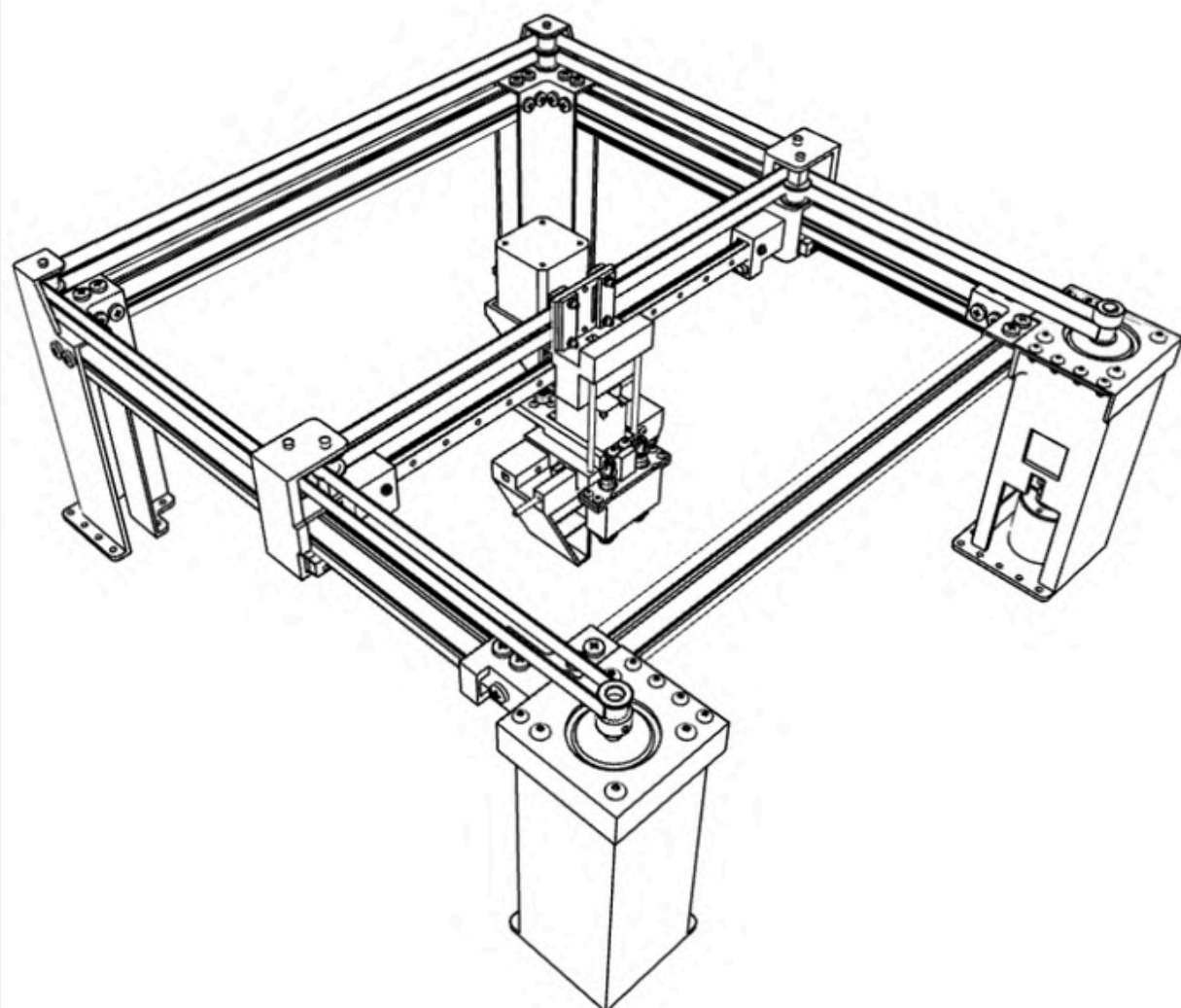
DRUK 3D



Filament	Model	Support	Purged	Tower	Total
■ 1	555.24 m 1615.98 g	94.28 m 274.39 g	0.09 m 0.26 g	0.21 m 0.61 g	649.82 m 1891.24 g
■ 2	59.46 m 173.06 g	3.33 m 9.69 g	0.13 m 0.37 g	0.04 m 0.12 g	62.96 m 183.24 g
Total	614.71 m 1789.03 g	97.61 m 284.08 g	0.22 m 0.63 g	0.25 m 0.73 g	712.78 m 2074.47 g
Total cost: 86.09					
Time Estimation					
Plate 1	14h14m				
Plate 2	2h6m				
Plate 3	2h1m				
Plate 4	2h13m				
Plate 5	6h27m				
Plate 6	2h20m				
Plate 7	15m45s				
Plate 8	2h21m				
Plate 9	2h43m				
Plate 10	1h34m				
Plate 11	1h56m				
Plate 12	1h44m				
Plate 13	4h11m				
Plate 14	3h0m				
Plate 15	4h40m				
Plate 16	4h59m				
Plate 17	4h39m				
Plate 18	2h3m				
Plate 19	2h3m				
Plate 20	46m1s				
Plate 21	1h33m				
Total	2d19h50m				

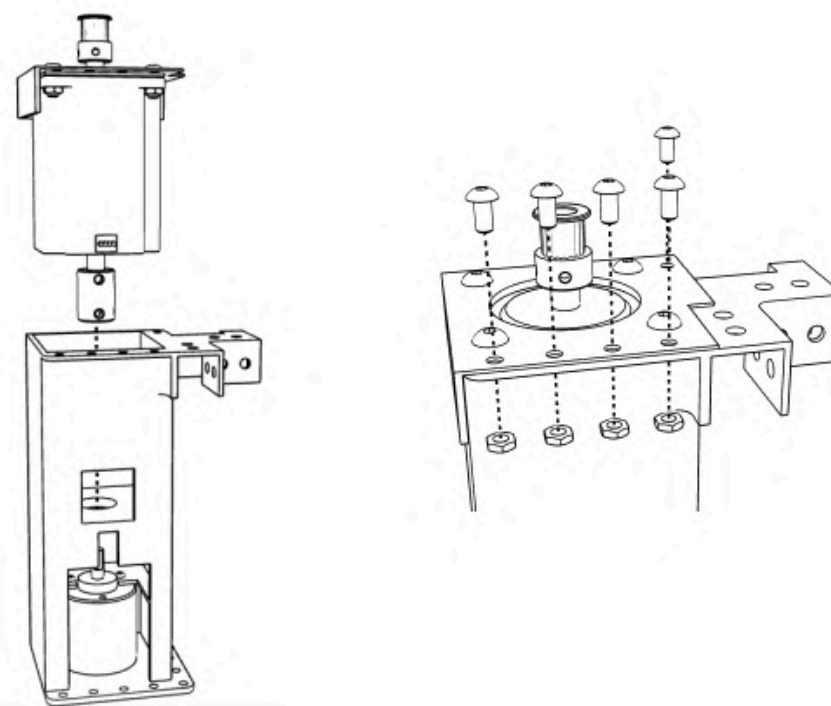
INSTRUKCJA

Instrukcja montażu CSAimBot

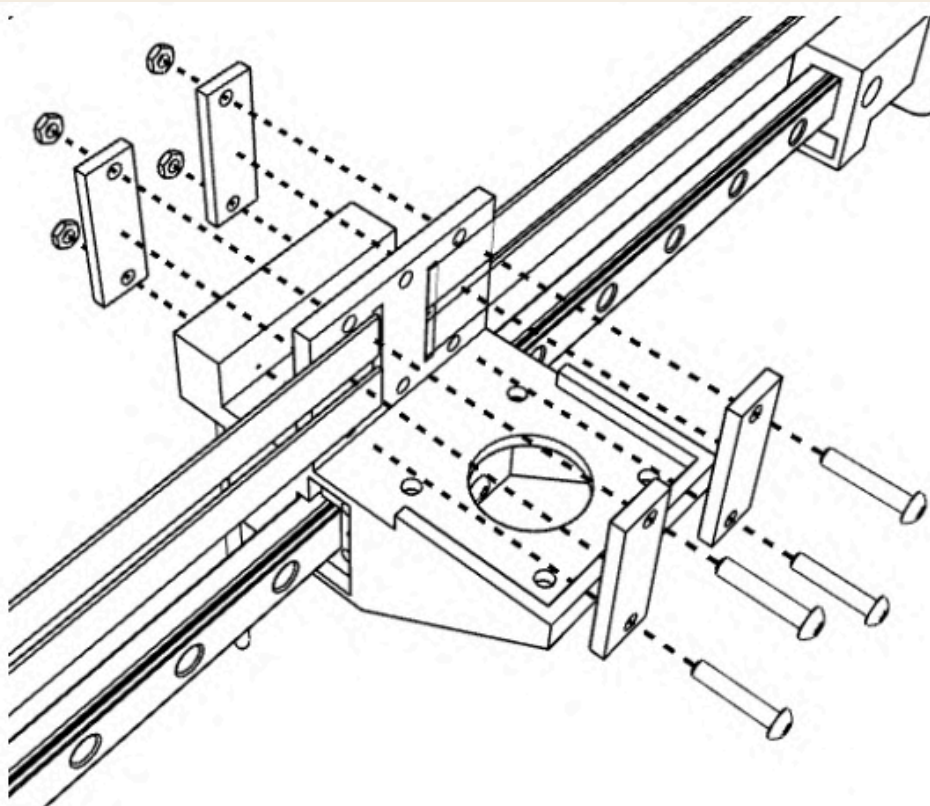


1

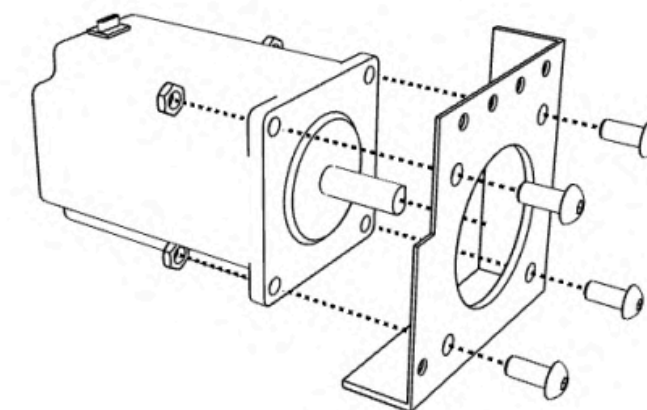
4



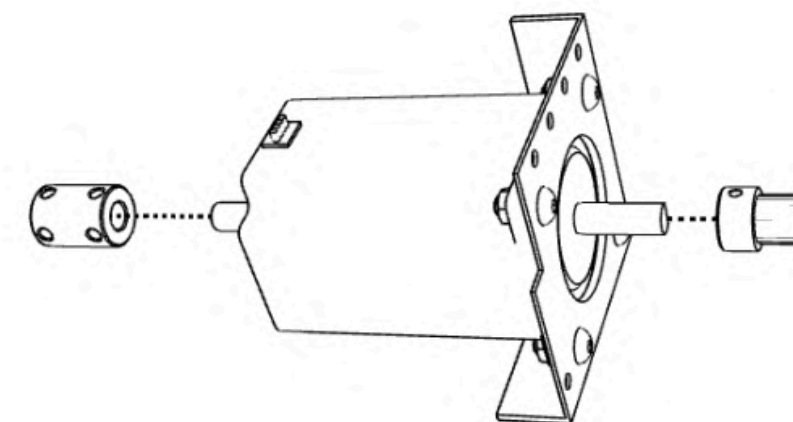
3



1



2



14. Spis Elementów:

Elementy złączne:

Śruba M3×6 mm



4x

Śruba M5×12 mm



8x

Nakrętka młotczkowa M5



42x

Śruba M3×10 mm



76x

Nakrętka sześciokątna M3



28x

Śruba M3×16 mm



2x

Nakrętka sześciokątna M5



8x

Śruba M5×10 mm



48x

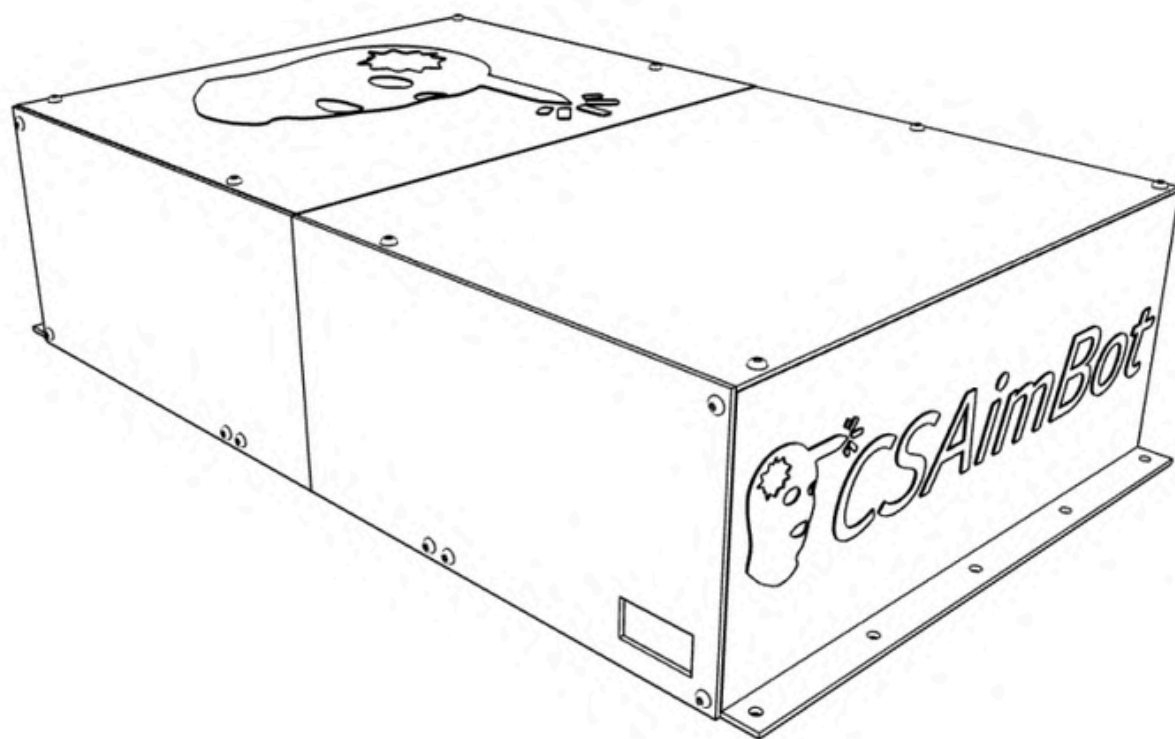
Nakrętka młotczkowa M3



32x

INSTRUKCJA

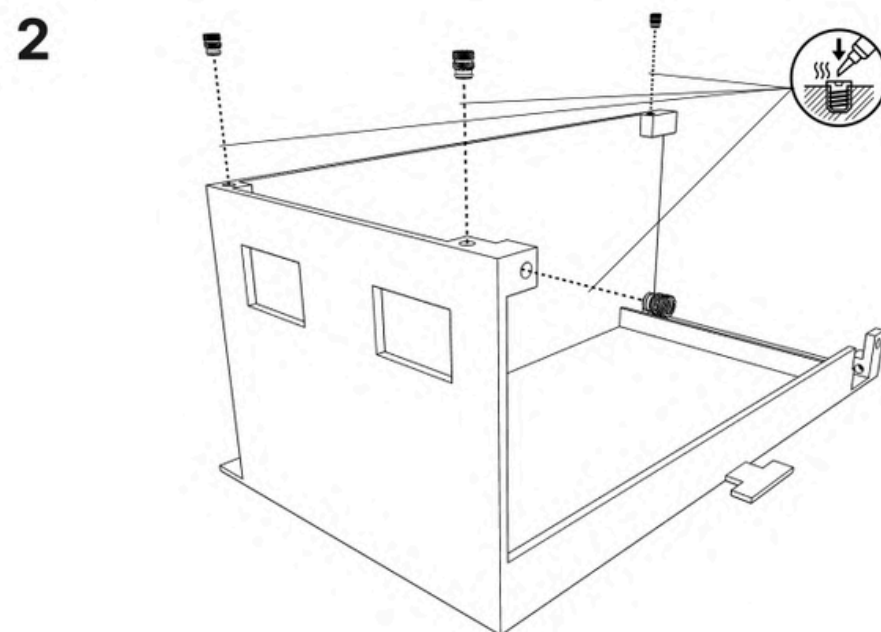
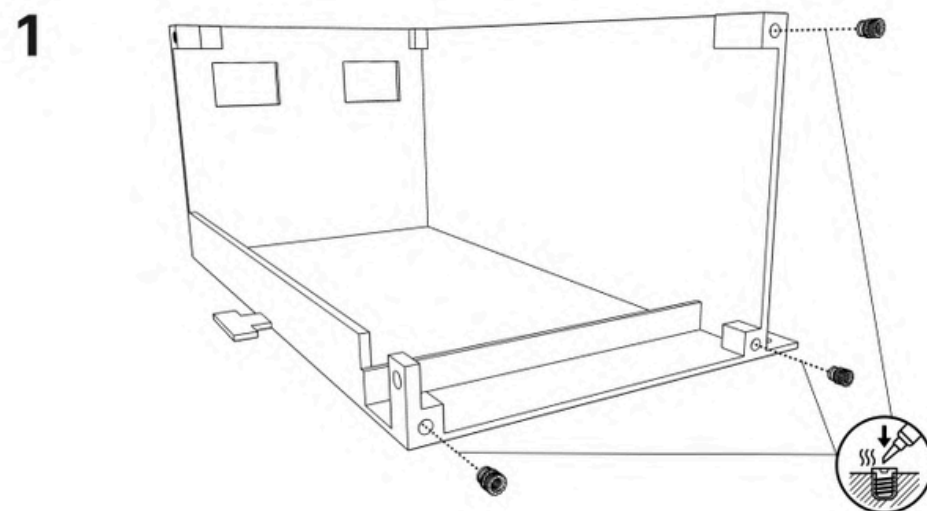
Instrukcja montażu Obudowy Elektroniki CSAimBot



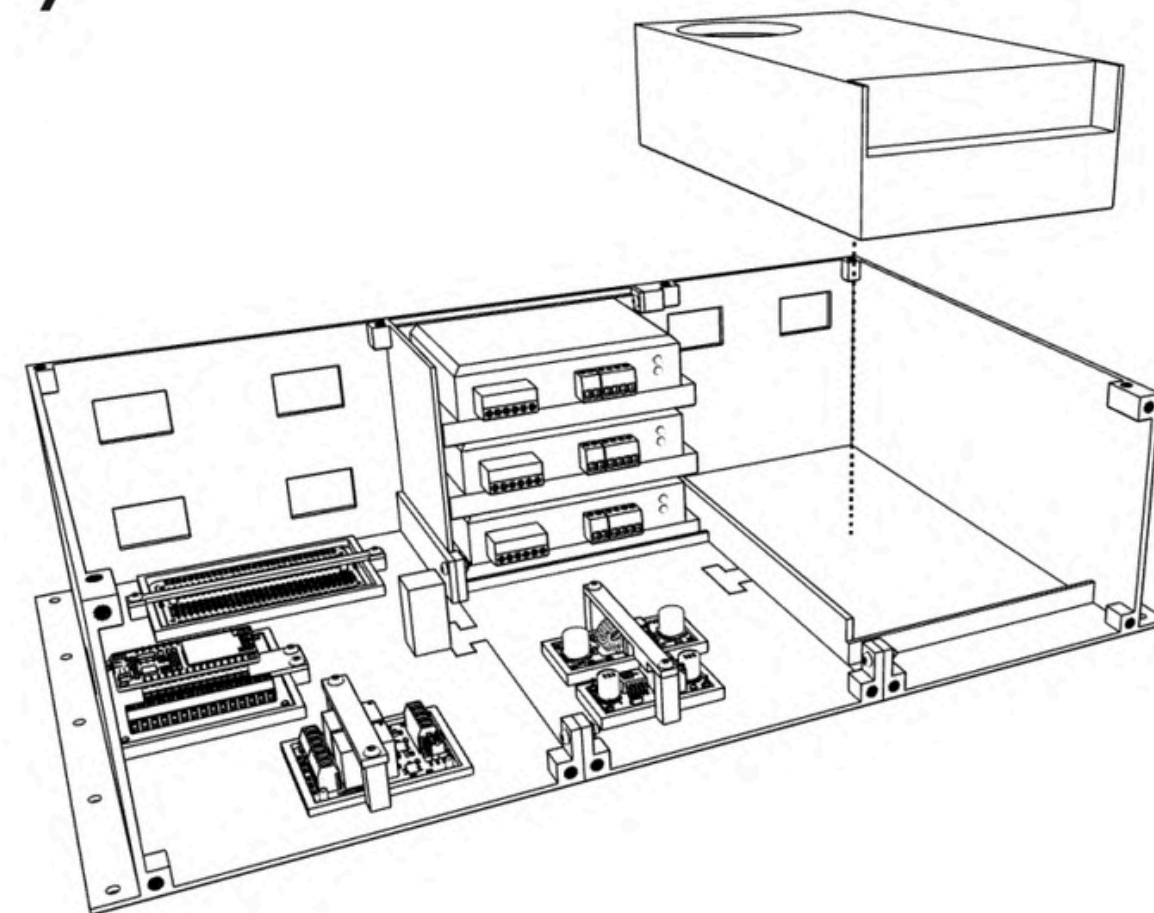
1. Moduł podstawy - strona prawa

Lista potrzebnych elementów:

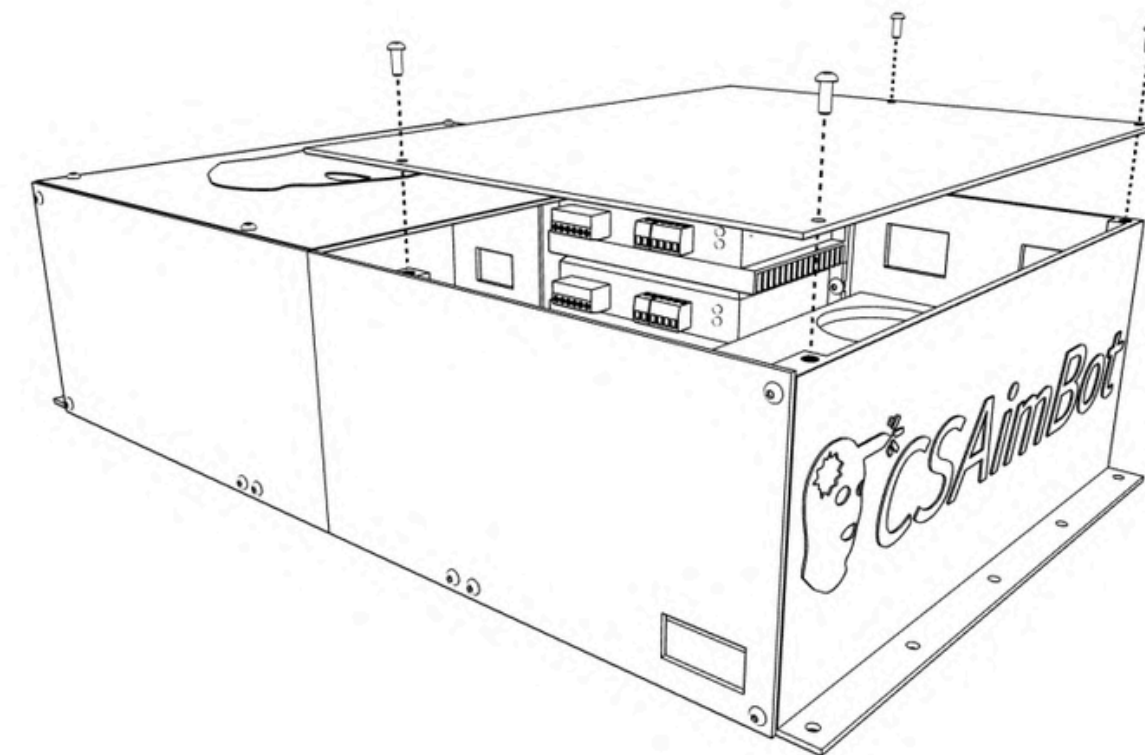
- Korpus podstawy - prawy x1
- Napis boczny x1
- Wkładka gwintowana M3×5mm x11



7



1

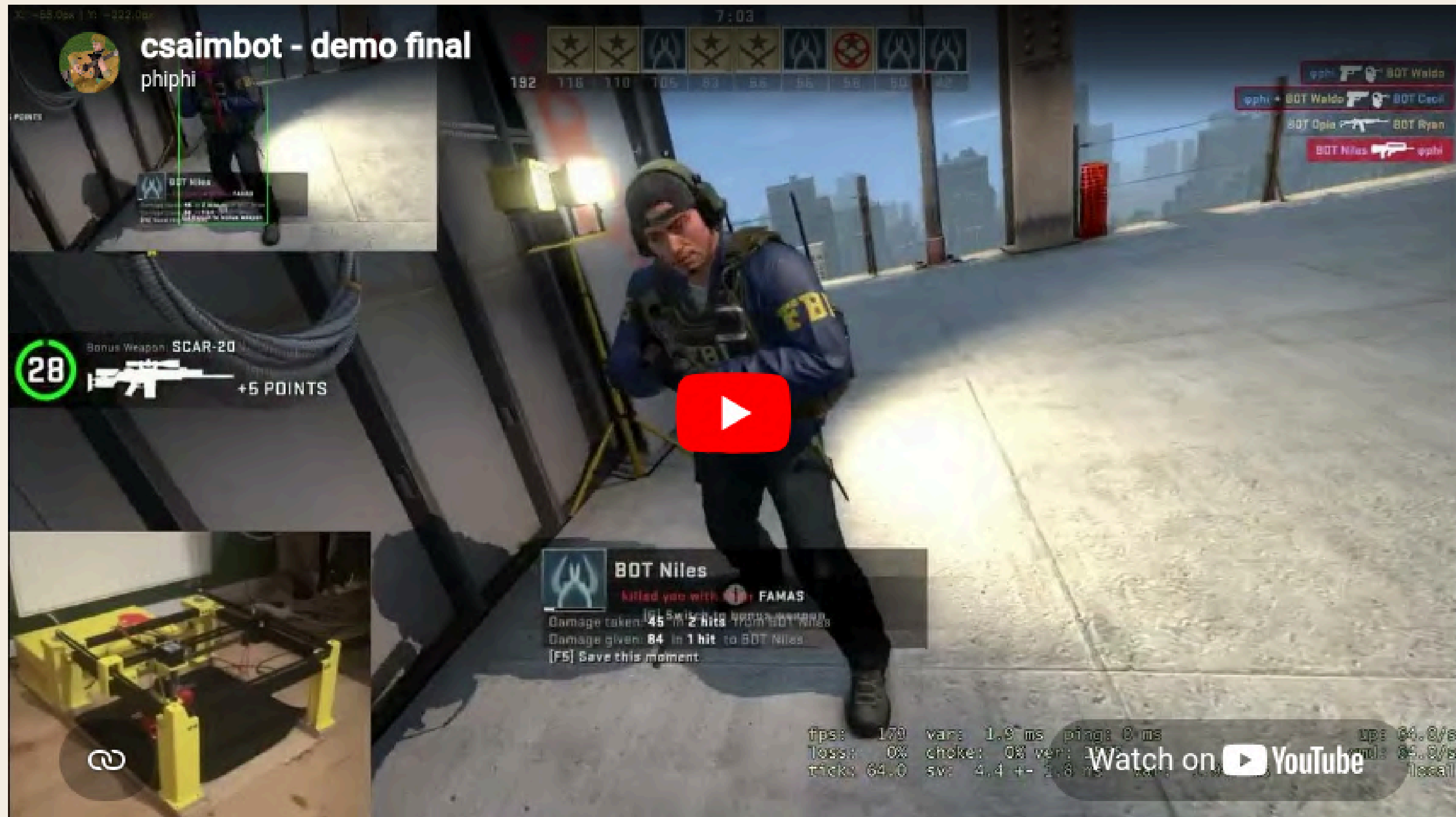


3

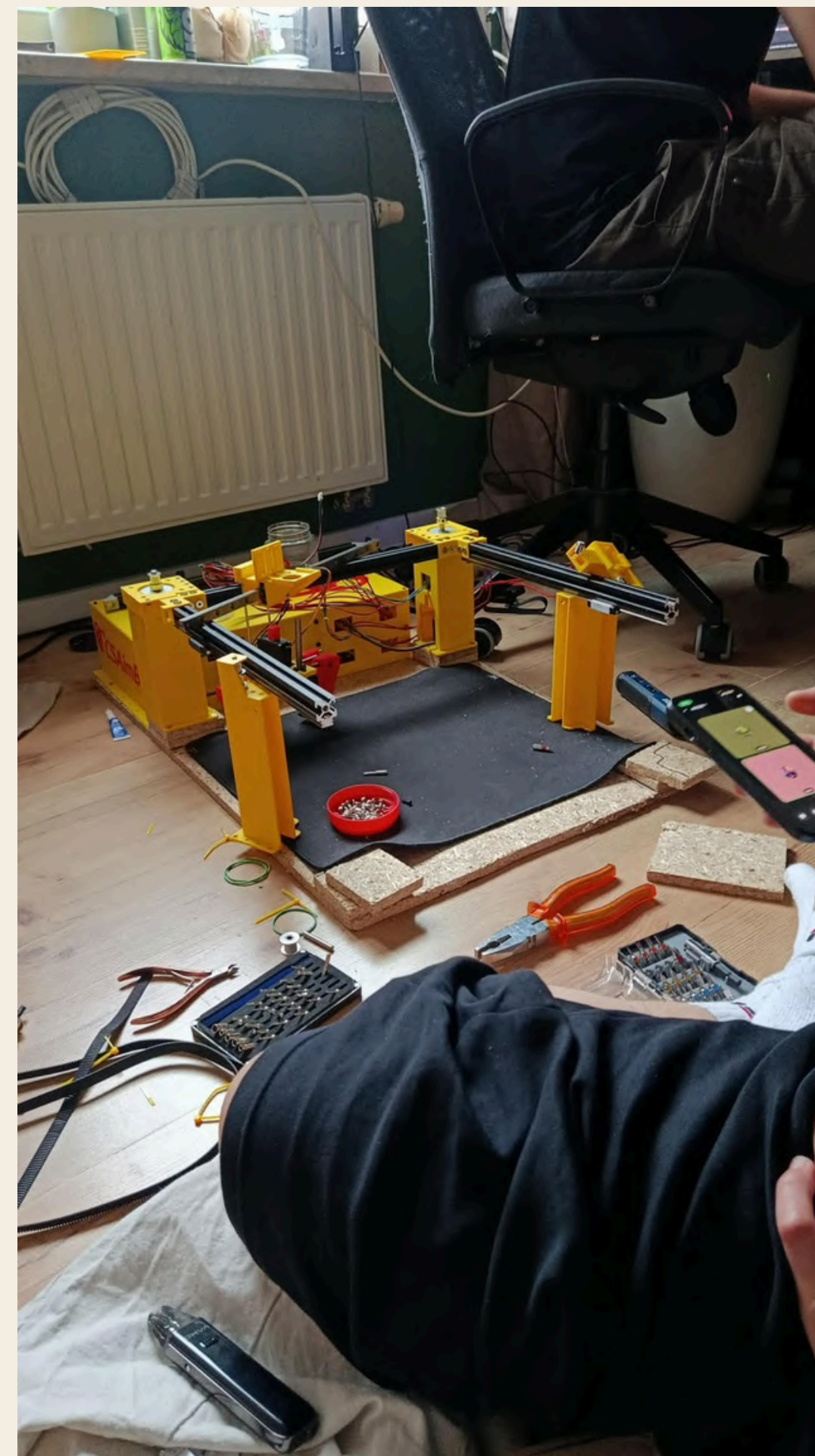
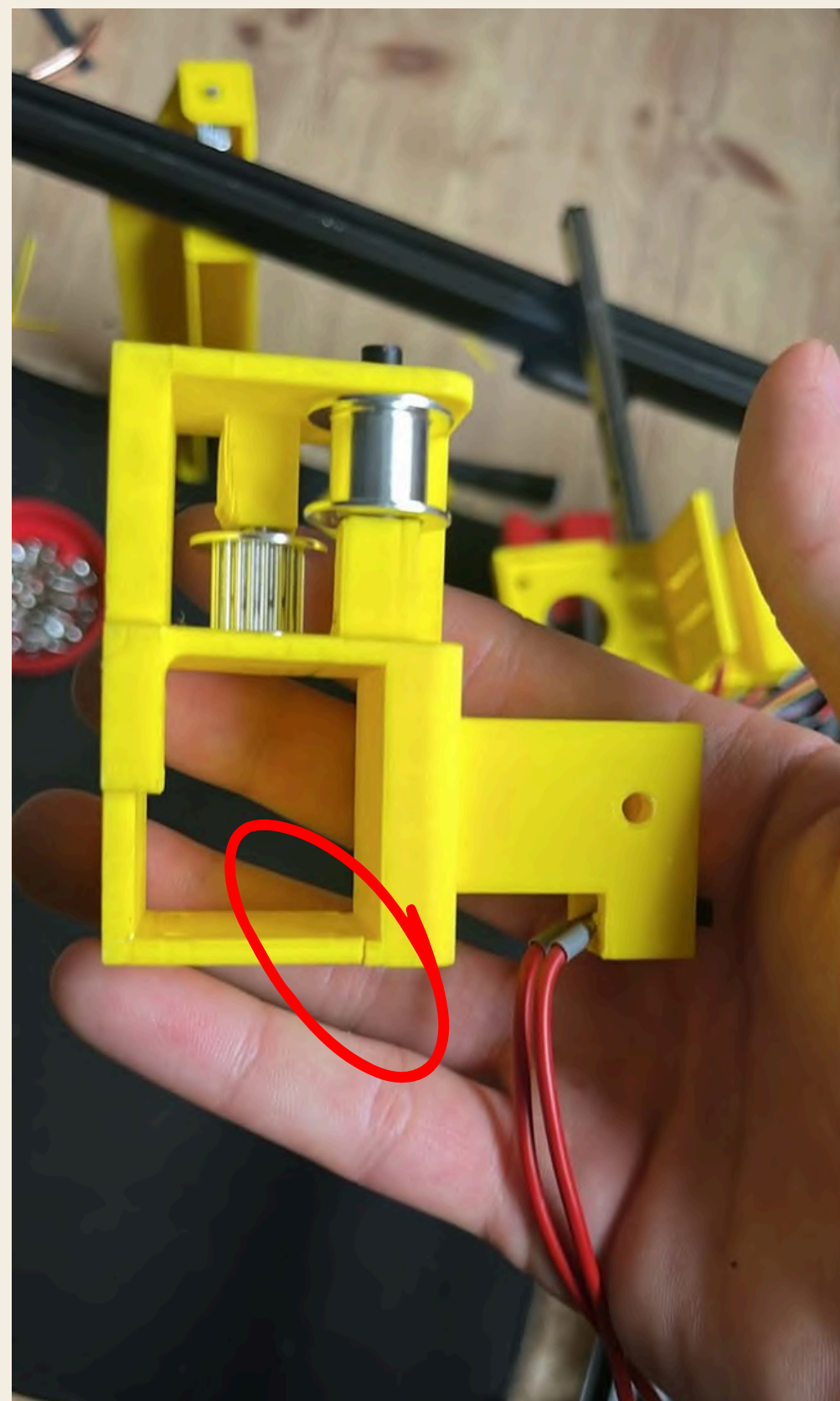
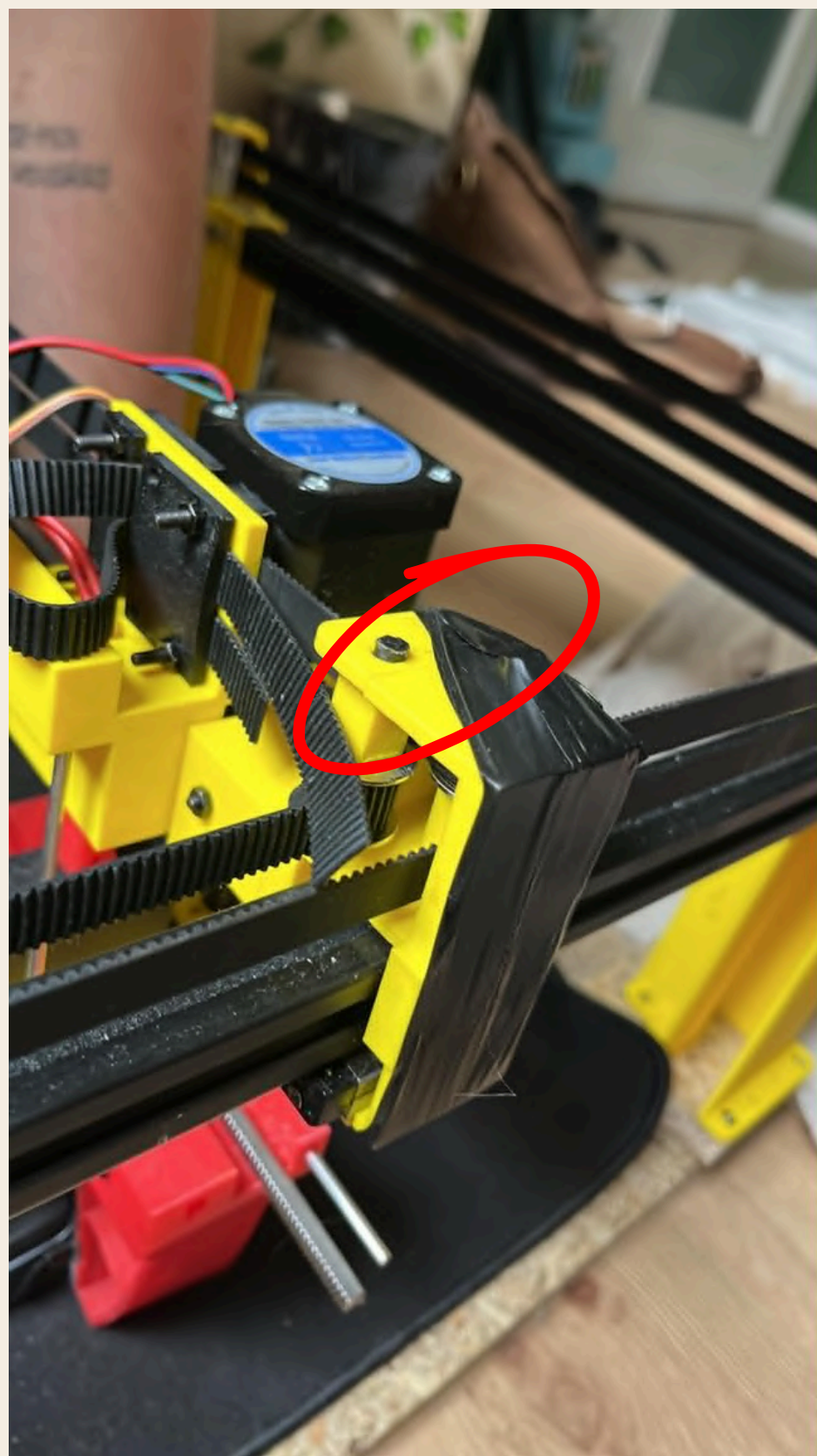
EFEKT KOŃCOWY



EFEKT KOŃCOWY



PROBLEMY



POKAZ NA ŻYWO

10 Czerwca 2026

DZIĘKUJEMY



Prezentowane przez
Filip Pietrzak, Tomasz Nazar